Supplementary Material

Identifiability of species interactions in a living material: Bayesian calibration of an extended Hamilton principle multiphase model

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S1 Model derivation

State variables and kinematics

Following Klempt et al. (2026, 2024), the biofilm is described at a material point by two sets of internal variables: the volume fraction $\phi_i \in (0,1)$ occupied by species i ($i = 1, \dots, n$; here $n = 5$), and the viability fraction $\psi_i \in (0,1)$ representing the proportion of living cells within species i. The living volume fraction is then

$$\bar{\phi}_i := \phi_i \, \psi_i, \qquad i = 1, \dots, n. \tag{S1}$$

An additional void fraction ϕ_0 closes the system via the holonomic constraint

$$\sum_{l=0}^{n} \phi_l(t) - 1 = 0 \quad \forall \, t. \tag{S2}$$

Variational formulation

The evolution equations result from the stationarity of the rate functional (Junker et al., 2014; Hackl, 1997), the local form of the extended Hamilton principle (Junker and Balzani, 2021; Junker et al., 2025):

$$\dot{\Psi} + \Delta + C \to \mathrm{stat}_{(\dot{\boldsymbol{\xi}}, \gamma)}, \tag{S3}$$

where $\boldsymbol{\xi} = (\boldsymbol{\phi}, \boldsymbol{\psi})^T$ collects the volume fractions and viability fractions, γ is a Lagrange multiplier enforcing the volume constraint, C is the constraint function, and Δ the dissipation function. The free energy density encodes nutrient-driven growth and antibiotic-induced killing:

$$\Psi = -\tfrac{1}{2} \, c^* \, \bar{\boldsymbol{\phi}}^T \mathbf{A} \, \bar{\boldsymbol{\phi}} + \tfrac{1}{2} \, \alpha^* \, \boldsymbol{\psi}^T \mathbf{B} \, \boldsymbol{\psi}, \tag{S4}$$

where c^* is the nutrient concentration, α^* the antibiotic concentration, $\mathbf{A} \in \mathbb{R}^{n \times n}$ the symmetric growth coefficient matrix, and $\mathbf{B} = \mathrm{diag}(b_1, \dots, b_n)$ the antibiotic sensitivity matrix.

The dissipation function is derived from the rate-dependent potential (Klempt et al., 2026):

$$\Delta = \tfrac{1}{2} \, \dot{\bar{\boldsymbol{\phi}}}^T \boldsymbol{\eta} \, \dot{\bar{\boldsymbol{\phi}}} + \tfrac{1}{2} \, \dot{\boldsymbol{\phi}}^T \boldsymbol{\eta} \, \dot{\boldsymbol{\phi}}, \tag{S5}$$

with the diagonal viscosity matrix $\boldsymbol{\eta} = \mathrm{diag}(\eta_1, \dots, \eta_n)$. The first term couples $\dot{\phi}_i$ and $\dot{\psi}_i$ through $\dot{\bar{\phi}}_i = \dot{\phi}_i \psi_i + \phi_i \dot{\psi}_i$, giving rise to non-linear dissipative interactions; the second term regularises the system to full rank. Following Klempt et al. (2026), the viscosity parameters are set to $\eta_i = 1$ for all species, rendering them dimensionless scaling factors; the physical time scale is absorbed into the time-step size Δt.

The volume constraint (S2) enters through a Lagrange multiplier γ:

$$C = \gamma \Big(\sum_{l=0}^{n} \phi_l - 1 \Big). \tag{S6}$$

Stationarity conditions

Evaluating the stationarity conditions yields the coupled evolution equations (Klempt et al., 2026):

Volume fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \psi_i \, (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \eta_i(\dot{\phi}_i \, \psi_i^2 + \bar{\phi}_i \, \dot{\psi}_i + \dot{\phi}_i) + \gamma, \tag{S7}$$

Viability fraction ($i = 1, \ldots, n$):

$$0 = -c^* \, \phi_i \, (\mathbf{A}\bar{\boldsymbol{\phi}})_i + \alpha^* \, b_i \, \psi_i + \eta_i(\dot{\psi}_i \, \phi_i^2 + \bar{\phi}_i \, \dot{\phi}_i) + \gamma, \tag{S8}$$

Void fraction:

$$0 = \gamma + \dot{\phi}_0, \tag{S9}$$

Volume constraint:

$$0 = \sum_{l=0}^{n} \phi_l - 1, \tag{S10}$$

In the numerical implementation, a barrier penalty $\Psi_{\text{bar}} = K_p[\sum_{i=0}^{n} \phi_i^{-2}(1-\phi_i)^{-2} + \sum_{i=1}^{n} \psi_i^{-2}(1-\psi_i)^{-2}]$ with $K_p = \eta_i \times 10^{-4}$, which diverges at 0 and 1, is added to enforce the box constraints $\phi_i, \psi_i \in (0, 1)$ (Klempt et al., 2026); its contribution is omitted from Eqs. (S7)–(S9) for clarity. The initial state is the day-1 composition scaled to $\sum_i \phi_i(0) = 1 - 10^{-6}$, so that the void fraction starts at $\phi_0(0) = 1 - \sum_i \phi_i(0) \approx 0$, and all species start fully viable, $\psi_i(0) = 0.999$; the barrier keeps the state inside $(0, 1)$ from the first time step on (e.g. $\phi_0 \approx 3 \times 10^{-9}$ after one step). The system (S7)–(S10) constitutes $2n + 2$ coupled non-linear equations for the unknowns $(\phi_1, \ldots, \phi_n, \phi_0, \psi_1, \ldots, \psi_n, \gamma)$, solved at each time step via Newton's method with an implicit Euler discretisation ($\Delta t = 10^{-4}$, $N_{\text{step}} = 2{,}500$ steps, $t_{\max} = 0.25$; day d is mapped to $t = 0.2375\, d/21$, so day 21 lies at 95% of the integration window).

Only the symmetric part of $\mathbf{A}$ enters the quadratic free energy, so $\mathbf{A}$ is taken symmetric, $a_{ij} = a_{ji}$, and the interactions derive from this potential; the variational derivation ensures thermodynamic consistency (Junker and Balzani, 2021). Since c^* enters the evolution equations only through the product $c^* a_{ij}$, its absolute value is non-identifiable; we fix $c^* = 25$ so that the inferred a_{ij} remain $\mathcal{O}(1)$. The Heine et al. experiments do not involve antibiotic agents, so we set $\alpha^* = 0$, which eliminates the decay term $\alpha^* b_i \psi_i$ from Eq. (S8); the antibiotic sensitivity matrix $\mathbf{B}$ is therefore inactive and excluded from estimation. The symmetric interaction matrix $\mathbf{A}$ has 15 independent entries, with species numbered 1 So, 2 An, 3 Vei, 4 Fn, 5 Pg; these form the 15-dimensional parameter vector $\boldsymbol{\theta}$.

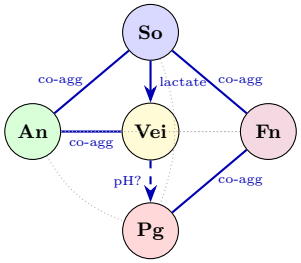
S2 Reported interactions among the five taxa

Figure S1 and Table S1 summarise interactions reported in the literature; they are used only to interpret the inferred coefficients, never as constraints.

S3 Likelihood weights and algorithm

S4 Prior boxes

Table S4 lists the uniform prior box of every interaction entry for each condition and stage. The stage-(i) boxes were set from earlier calibrations of a previous version of the model on the same data. Some of them exclude negative values or allow only small ones (CS: $a_{11} \in [0, 2]$; DH: $a_{11}, a_{22}, a_{44} \in [0, 5]$, $a_{12} \in [0.5, 3]$, $a_{23} \in [0, 3]$, $a_{45} \in [-0.5, 6]$), so in stages (i)–(ii) the signs of



Pair	Mechanism	Reported flow
So–An	Co-aggregation	–
So–Vei	Lactate exchange	So to Vei
An–Vei	Co-aggregation; Vei uses lactate of So and An	–
So–Fn	Co-aggregation (Fn as bridging organism)	–
Fn–Pg	Co-aggregation (Fn bridges to Pg)	–
Vei–Pg	pH rise from *Veillonella* lactate metabolism (hypothesised in Heine et al., 2025; *Veillonella*-conditioned medium did not change Pg growth)	Vei to Pg (hypothesised)

Table S1: Previously reported or biologically supported interactions (Kolenbrander et al., 2010; Heine et al., 2025) used to interpret the inferred coefficients. The model coefficients a_{ij} are symmetric; the last column refers to the reported metabolite flow (–: physical co-aggregation, no metabolite flow), not to the model. All 15 parameters are estimated; none is fixed a priori.

Figure S1: Interactions considered in interpreting the inferred coefficients, compiled from the literature (Kolenbrander et al., 2010; Heine et al., 2025). Solid: previously reported or biologically supported interaction (labels: reported mechanism; arrows: reported metabolite flow; lines without arrows: co-aggregation; neither is a property of the model, whose matrix $\mathbf{A}$ is symmetric). Dashed: hypothesised, not supported by the conditioned-medium experiment of Heine et al. (2025). Dotted grey: no interaction considered in the interpretation; this does not mean that none exists. The network is not used in the inference: all 15 entries of $\mathbf{A}$ are estimated with the same uniform prior box.

Weight	Value	Applies to
λ_{Pg}	5	Pg, all time points, both compositions
λ_{late}	3	Days 15 and 21, all species
λ_{rare}	0.1	species with mean fraction $< 5\%$
λ_{ψ}	3.0 (DH), 2.0 (DS), 1.5 (CS, CH)	viability channel, stages (iii)–(iv)

Table S2: Likelihood weights. λ_{Pg}, λ_{late} and λ_{rare} act on both compositions (viable-cell fractions from PMA-qRT-PCR and total-count fractions without PMA); the viability channel is the community live fraction from SYTO9/propidium iodide staining.

Metric	CPU (12 cores)	GPU (RTX 4090)	Speedup
Matched benchmark (DH, $N_p = 500$, $n_{\mathrm{mut}} = 10$, $K = 9$ tempering steps):			
Per tempering step	2 400 s	14 s	170×
Total	21 600 s	147 s	**147**×

Table S3: Computational cost comparison. The matched benchmark (500 particles, identical settings) isolates the GPU speed-up at about 150× (170× per tempering step, 147× end to end). The cost of every inference stage of the paper is listed in Table S14.

Algorithm S1 Staged GPU-accelerated TMCMC (one seed; every stage is run with three seeds and checked as in the main text)

Require: Composition data $\mathbf{y}_{\text{obs}}$, viability data, N_p particles, GPU device
Ensure: MAP estimate $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, log-evidence $\ln \hat{Z}$

1: ▷ **Stage (i): viable and total-count composition, ψ_i fixed to the data, initial box**
2: Draw $\{\boldsymbol{\theta}_j\}_{j=1}^{N_p}$ from the stage prior; set $\beta_0 \leftarrow 0$
3: **while** $\beta_m < 1$ **do**
4: Find β_{m+1} s.t. $\mathrm{CoV}[\{w_j\}] = \delta_{\text{target}}$ via bisection
5: Compute weights $w_j \leftarrow p(\mathbf{y}_{\text{obs}} \mid \boldsymbol{\theta}_j)^{\beta_{m+1}-\beta_m}$
6: Resample N_p particles $\propto w_j$ (systematic)
7: Estimate weighted covariance $\hat{\boldsymbol{\Sigma}}^{(m)}$ (weighted particle covariance)
8: **for** $k = 1$ **to** n_{mut} **do**
9: Propose $\boldsymbol{\theta}'_j$ for all j (random-walk or differential-evolution move)
10: Evaluate all N_p likelihoods in parallel via `jax.vmap` on GPU
11: Accept or reject (Metropolis–Hastings); reject proposals outside the box
12: **end for**
13: **end while**
14: $K \leftarrow m$ (number of tempering steps)
15: Record MAP and posterior standard deviation (SD) per parameter
16: ▷ **Stage (ii): composition likelihood, box mean ± 4 SD**
17: Repeat lines 2–14 on the narrowed box, starting from the previous particles
18: ▷ **Stages (iii)–(iv): add the viability channel; initial box, then mean ± 4 SD**
19: Evolve ψ_i and add the weighted live-fraction channel (Table S2)
20: Repeat lines 2–14 for each stage
21: **return** $\hat{\boldsymbol{\theta}}^{\text{MAP}}$, posterior samples, $\ln \hat{Z}$

these entries are partly set by the prior; all of these bounds were widened in stage (iii). An entry was widened where more than 20% of its posterior mass lay in the outer 5% of the range on one side, or where this fraction grew by more than 0.10 from the previous stage; the stage was then repeated. The last column gives $\ln V = \sum_k \ln(u_k - l_k)$, the log-volume of the box, which enters $\ln \hat{Z}$ and is the reason why $\ln \hat{Z}$ is not compared across conditions.

S5 Runs and convergence

Table S5 lists, for every condition and stage, the number of particles and mutation steps, the number of tempering steps, the acceptance rate, the moves per parameter, and the spread over the three seeds of the maximum log-likelihood and of the posterior medians (criteria (c)–(e) of the main text). Criteria (a) and (b) are met by all runs. Criterion (c) counts accepted moves rather than the acceptance rate: with optimally scaled random-walk proposals (Roberts and Rosenthal, 2001), $5d$ accepted moves displace each coordinate by about five posterior standard deviations, independently of the dimension d. To check that the warm start does not bias stage (iii), one CH stage-(iii) run was started with all particles drawn from the prior, on the same box as the three warm-started seeds: its maximum log-likelihood (-26.74) lies within their range (-26.71 to -27.25), and the medians and 90% bounds of a_{33} and a_{45} lie within 0.03 of the range spanned by the three warm-started seeds. Stages (ii)–(iv) of CS and CH, stages (iii)–(iv) of DS and stages (ii)–(iii) of DH meet all criteria (largest median spread 0.09 posterior SD in CS and CH, 0.16 in DS, 0.13 in DH). Stage (ii) of DS fails (d): the best-known maximum log-likelihoods of the seeds (-59.07, -60.74, -60.77; the first from top-10 polishing of one seed) spread by 1.70; stage (iii), half of whose particles start from the prior, does not inherit this (spread 0.11). Stage

Table S4: Prior boxes per condition and stage for the 15 free entries. (i) initial box; (ii) stage-(i) posterior mean ± 4 SD; (iii) back on the box of stage (i), with the bounds widened (bold) where more than 20% of the posterior mass lay in the outer 5% on one side; (iv) stage-(iii) posterior mean ± 4 SD, clipped to the stage-(iii) box (seed with the highest log-likelihood). $\ln V = \sum_k \ln(u_k - l_k)$ over the free entries.

CS

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[0, 2]	[0, 1.88]	**[-10, 2]**	[-10, 2]
a_{12}	[-0.5, 2]	[0.75, 2]	[-0.5, 2]	[-0.5, 2]
a_{22}	[-1, 3]	[-1, 3]	[-1, 3]	[-0.93, 3]
a_{33}	[-1, 1.5]	[-1, -0]	**[-15, 20]**	[-15, 0.85]
a_{34}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{44}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{13}	[-1, 3]	[1.58, 3]	**[-1, 20]**	[0.35, 16.3]
a_{14}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{23}	[-1, 3]	[-0.12, 2.69]	**[-5, 3]**	[-3.74, 3]
a_{24}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{55}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{15}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{25}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{35}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{45}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
$\ln V$	12.9	9.9	26.3	25.0

CH

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[-1, 1.5]	[-0.57, 1.5]	**[-5, 1.5]**	[-2, 0.42]
a_{12}	[-1, 2.5]	[-0.22, 2.5]	**[-5, 2.5]**	[-1.41, 0.79]
a_{22}	[-1, 2.5]	[0.64, 2.5]	[-1, 2.5]	[-1, 1.46]
a_{33}	[-1, 1.5]	[-1, 1.5]	**[-15, 20]**	[-9.32, 3.7]
a_{34}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{44}	[-1, 1.5]	[-1, 1.5]	**[-1, 5]**	[-1, 5]
a_{13}	[-1, 3]	[-0.03, 3]	[-1, 3]	[-1, 3]
a_{14}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{23}	[-1, 3]	[-0.97, 3]	[-1, 3]	[-1, 3]
a_{24}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{55}	[-1, 1.5]	[-1, 1.5]	**[-5, 1.5]**	[-5, 1.5]
a_{15}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{25}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{35}	[-1, 1]	[-1, 1]	**[-5, 1]**	[-5, 1]
a_{45}	[-1, 2]	[-1, 2]	[-1, 2]	[-1, 2]
$\ln V$	14.2	12.8	27.0	23.4

DS

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[-1, 2]	[-1, 2]	[-1, 2]	[-1, 2]
a_{12}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{22}	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]	[-1, 1.5]
a_{33}	[-15, 20]	[0.06, 3.86]	[-15, 20]	[-1.2, -0.07]
a_{34}	[-15, 20]	[0.13, 5.78]	[-15, 20]	[0.2, 2.6]
a_{44}	[-1, 2.5]	[-1, 0.74]	**[-5, 2.5]**	[-5, -1.22]
a_{13}	[-1, 1.5]	[-1, 1.5]	**[-5, 1.5]**	[-5, -0.47]
a_{14}	[-1, 1]	[-1, 1]	[-1, 1]	[-1, 1]
a_{23}	[-1, 3]	[-1, 3]	**[-5, 3]**	[-3.11, 0.73]
a_{24}	[-0.5, 3]	[-0.5, 3]	[-0.5, 3]	[-0.5, 3]
a_{55}	[-1, 4]	[-1, 4]	**[-5, 4]**	[-5, 3.46]
a_{15}	[-0.5, 2.5]	[-0.5, 2.5]	**[-5, 2.5]**	[-5, 2.5]
a_{25}	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]	[0.1, 2.5]
a_{35}	[-3, 3]	[-0.37, 3]	[-3, 3]	[-0.84, 0.72]
a_{45}	[-15, 20]	[-0.87, 7.27]	[-15, 20]	[-2.27, 3.99]
$\ln V$	24.5	17.7	28.4	17.1

DH

Entry	(i)	(ii)	(iii)	(iv)
a_{11}	[0, 5]	[0, 5]	**[-10, 5]**	[-10, 5]
a_{12}	[0.5, 3]	[0.5, 3]	**[-10, 3]**	[-10, 3]
a_{22}	[0, 5]	[1, 4.29]	**[-10, 5]**	[-10, 5]
a_{33}	[-1, 3]	[-0.56, 1.75]	**[-15, 20]**	[-0.9, -0.07]
a_{34}	[-1, 1]	[-1, 1]	**[-15, 20]**	[-2.6, 1.49]
a_{44}	[0, 5]	[0, 5]	**[-10, 5]**	[-10, 5]
a_{13}	[-3, 3]	[-3, 2.67]	**[-8, 3]**	[-8, 3]
a_{14}	[-1.5, 3]	[-1.5, 3]	**[-8, 3]**	[-8, 3]
a_{23}	[0, 3]	[0, 1.89]	**[-15, 20]**	[-1.04, 1.9]
a_{24}	[-0.5, 3]	[1.36, 3]	**[-15, 20]**	[-4.23, 14.2]
a_{55}	[-1.5, 5]	[-1.5, 5]	[-1.5, 5]	[-1.5, 5]
a_{15}	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]	[-0.5, 2.5]
a_{25}	[-0.5, 2.5]	[-0.5, 2.5]	**[-5, 2.5]**	[-5, 2.5]
a_{35}	[-0.5, 5]	[-0.5, 1.12]	**[-25, 20]**	[-25, 18.16]
a_{45}	[-0.5, 6]	[-0.5, 6]	**[-5, 20]**	[-5, 17.31]
$\ln V$	21.1	17.7	41.7	32.5

(iv) of DH meets (a)–(d) but not (e): the seed medians differ by 0.79 posterior SD for a_{22} and 0.69 for a_{24} and a_{13}, with 12.6–13.0 moves per free parameter, so the seeds settle in different modes rather than mixing too little; the final stage of DH is therefore stage (iii) (main text, Section 4.5). Stage (i) is exploratory: no condition meets (c) (2.7–4.2 moves per free parameter). Criterion (e) is stricter than common practice. Treating the three seeds as chains, the potential scale reduction factor $\hat{R}$ (Gelman and Rubin, 1992) of the identified entries is at most 1.001 for CS and CH (stages (ii)–(iv)) 1.003 for DS (stage (iv)) and 1.003 for DH (stage (iii)), and 1.11 for the DH stage-(iv) group that fails (e). Here criterion (e) and the conventional threshold $\hat{R} < 1.1$ agree; the recommended $\hat{R} < 1.01$ (Vehtari et al., 2021) is met by every accepted group. Values from `tools/rhat_seeds.py`. For CS and CH, stage (ii) was started from stage-(i) runs with 120 mutation steps that are not archived; the stage-(i) runs listed here (80 steps) give stage-(ii) bounds within 0.17 of those used.

Table S5: Runs of the final pipeline (three seeds per row): particles N_p, mutation steps per tempering step n_{mut}, tempering steps K, acceptance rate, moves per free parameter (criterion (c)), maximum log-likelihood (best seed, after local optimisation from the best particles where this was done; spread over the three seeds in parentheses, criterion (d)), largest spread of the seed medians over the identified entries in units of the pooled posterior SD (criterion (e), no mode splitting) and wall-clock hours per seed. DH: the final stage is (iii); stage (iv) fails (e) and is listed for completeness.

Cond.	Stage	N_p	n_{mut}	K	Accept.	Moves/dim	max log L (spread)	Med. spread	Hours
CS	(i)	1000	40	8	0.14–0.14	2.9–3.0	-4.15 (0.11)	0.20	1.6–1.8
CS	(ii)	2000	120	6	0.15–0.15	7.0–7.1	-3.87 (0.00)	0.06	3.5–4.0
CS	(iii)	2000	160	7–8	0.13–0.14	10.0–11.5	-33.00 (0.04)	0.07	7.9–8.8
CS	(iv)	5000	150	8	0.13–0.13	10.4–10.5	-32.34 (0.68)	0.09	7.1–8.1
CH	(i)	1000	40	8	0.13–0.13	2.7–2.8	-1.96 (0.13)	0.14	1.8–1.8
CH	(ii)	2000	150	6	0.14–0.14	8.3–8.4	-1.71 (0.02)	0.07	4.8–4.9
CH	(iii)	2000	150	8–9	0.13–0.13	10.7–12.0	-25.86 (0.01)	0.04	6.0–6.8
CH	(iv)	5000	150	9	0.13–0.14	12.0–12.5	-25.87 (0.18)	0.05	7.8–8.7
DS	(i)	1000	40	11–12	0.13–0.14	4.1–4.2	-63.37 (0.54)	0.34	2.9–3.2
DS	(ii)	3000	100	8–9	0.14–0.14	7.4–8.2	-59.07 (1.70)	0.15	6.1–7.0
DS	(iii)	18000	150	13–14	0.12–0.12	15.8–17.0	-96.46 (0.23)	0.06	21.4–25.1
DS	(iv)	18000	150	11	0.13–0.13	13.8–14.1	-96.28 (0.15)	0.16	17.1–18.9
DH	(i)	1000	40	10–11	0.12–0.12	3.2–3.6	-14.44 (0.90)	0.50	2.7–2.9
DH	(ii)	2000	80	8–9	0.13–0.13	5.7–6.5	-12.58 (0.35)	0.41	4.7–5.3
DH	(iii)	20000	120	12–13	0.11–0.11	10.8–11.4	-50.62 (0.09)	0.13	21.1–22.7
DH	(iv)	20000	150	11	0.11–0.12	12.6–13.0	-50.62 (0.92)	0.79	21.7–22.6

S6 Identifiability of the interaction entries

Table S7 lists, for the final stage (iv) of every condition (stage (iii) for DH), the median and the 90% interval of each entry, pooled over the three seeds. The reference box is the prior box of that stage (Table S4, column (iv); for DS and DH column (iii)). An entry whose pooled posterior SD is at least 0.8 times that of a uniform distribution on this box is shown in grey and excluded from the seed-agreement criterion: the data do not constrain its magnitude beyond the prior box. Among these, entries whose 90% interval excludes zero are marked s.o. (sign only), the others n.i. (not identified). All other entries are called identified (constrained by the data); an identified entry need not be sign-determined (e.g. a_{45} in DS and DH). Where the stage-(iv) box was narrowed to the stage-(iii) mean ± 4 SD, the uniform SD on that box is about 2.3 stage-(iii) SDs; for these entries the classification is in effect made against the stage-(iii) box.

The threshold 0.8 is a convention. Table S6 counts the identified entries for thresholds between 0.5 and 0.9. The counts change, most in the commensal conditions, where some entries lie close to 0.8 (CS: 0.77 and 0.81; CH: 0.73 and 0.83), whereas DS has no entry between 0.71 and 0.93. The pattern does not change: at every threshold the dysbiotic conditions have at least as many identified entries as the commensal ones, and the entries identified in the commensal conditions are interactions of So, An and Vei; at 0.9 they also include So–Fn and So–Pg, the two entries that are sign-determined in CH.

Curvature of the likelihood. As a check that does not use the posterior samples, we computed the Hessian $\mathbf{H}$ of $-\ln p(\mathbf{y}_{\mathrm{obs}} \mid \boldsymbol{\theta})$ at the MAP over the 15 free entries, scaled each entry by the standard deviation of a uniform distribution on its box, and counted the eigen-directions with scaled curvature of at least $1/0.8^2$, i.e. directions in which the likelihood is narrower than the identification threshold. Their number is 5 in CS, 6 in CH, 6 in DS and 7 in DH, against 5, 6, 9 and 8 entries identified from the posterior. The counts agree in CS and CH; in DS and DH the posterior identifies more entries (three and one) than the local curvature, because entries can be constrained jointly along correlated directions and by the non-quadratic shape of the

Table S6: Number of identified entries of **A** (of 15) for different thresholds on the ratio of the pooled posterior SD to the SD of a uniform distribution on the prior box; *value used throughout. DH: stage (iii). Generated by `tools/ident_threshold_table.py`.

Threshold	0.5	0.7	0.8*	0.9
CS	3	5	5	6
CH	3	6	6	6
DS	6	9	9	10
DH	7	8	8	10

Table S7: Posterior median [5%, 95%] of the 15 entries at stage (iv) (stage (iii) for DH), three seeds pooled. Grey: the posterior spreads over most of the prior box of the stage (SD $\geq$ 0.8 times that of a uniform distribution on the box); n.i.: not identified, s.o.: sign only (the 90% interval excludes zero).

Entry	CS	CH	DS	DH
a_{11}	−4.20 [−7.0, −1.6]	−0.72 [−1.4, −0.4]	+0.38 [−0.9, +1.8] n.i.	−3.52 [−9.4, +4.0] n.i.
a_{12}	+1.27 [+0.0, +1.9] s.o.	−0.31 [−0.8, +0.1]	−0.07 [−0.9, +0.9] n.i.	−4.21 [−9.3, +1.6] n.i.
a_{22}	+1.44 [+0.5, +2.4]	−0.42 [−0.9, +0.5]	+0.16 [−0.9, +1.4] n.i.	−2.23 [−6.4, +0.2]
a_{33}	−11.07 [−14.6, −5.3]	−2.50 [−5.8, −0.6]	−0.64 [−0.9, −0.4]	−0.46 [−0.6, −0.4]
a_{34}	−2.21 [−4.7, +0.6] n.i.	−2.50 [−4.8, +0.5] n.i.	+1.45 [+0.9, +1.9]	−0.51 [−1.3, +0.2]
a_{44}	−0.01 [−0.9, +0.9] n.i.	+1.72 [−0.8, +4.7] n.i.	−4.29 [−4.9, −2.7]	−5.77 [−9.6, −0.2]
a_{13}	+8.45 [+4.7, +11.3]	+0.96 [+0.3, +2.4]	−4.22 [−4.9, −2.4]	−1.44 [−6.9, +1.1] n.i.
a_{14}	−1.89 [−4.7, +0.7] n.i.	−1.31 [−4.5, +0.6] n.i.	−0.06 [−0.9, +0.9] n.i.	−3.02 [−7.5, +2.2] n.i.
a_{23}	−0.12 [−1.4, +1.7]	+1.64 [+0.4, +2.7]	−1.29 [−2.0, −0.5]	+0.39 [−0.1, +1.0]
a_{24}	−2.10 [−4.7, +0.7] n.i.	−2.52 [−4.8, +0.5] n.i.	+1.62 [−0.1, +2.8] n.i.	+4.98 [+1.3, +8.8]
a_{55}	−0.03 [−0.9, +0.9] n.i.	−1.92 [−4.7, +1.1] n.i.	−1.08 [−3.6, +0.2]	+1.25 [−1.2, +4.5] n.i.
a_{15}	−2.00 [−4.7, +0.6] n.i.	−2.08 [−4.7, +0.7] n.i.	−2.34 [−4.8, +1.7] n.i.	+1.02 [−0.3, +2.3] n.i.
a_{25}	−2.18 [−4.7, +0.7] n.i.	−2.71 [−4.8, +0.4] n.i.	+2.11 [+1.1, +2.5]	−1.08 [−4.5, +2.0] n.i.
a_{35}	−2.08 [−4.7, +0.7] n.i.	−2.58 [−4.8, +0.5] n.i.	−0.06 [−0.3, +0.2]	−5.72 [−20.9, −0.6]
a_{45}	+0.01 [−0.9, +0.9] n.i.	+0.48 [−0.9, +1.8] n.i.	+0.63 [−0.2, +2.1]	+3.11 [−2.4, +9.2]

likelihood, which a single Hessian at the MAP does not capture. The Hessian is evaluated at the final MAP (locally optimised in CS and CH, the best particle in DH); it is not positive definite at any of them (two, one, one and four negative eigenvalues in CS, CH, DS and DH), so entry-wise Laplace intervals are not reported.

S7 Recovery test on synthetic data

To test whether the pipeline recovers known interaction coefficients, and whether the classification of Section S6 reflects the information in the data rather than the sampler, we generated synthetic data from a point of the CH posterior (the best particle of the stage-(iv) seed with the highest log-likelihood at the time, −26.52) at the measured time points, with Gaussian noise of the same standard deviations as in the likelihood (viable and total-count compositions and live fraction), and inferred them again with the stage-(iv) settings (box, weights, 5000 particles, 150 mutation steps) and three seeds. The log-likelihood of the true parameters on the synthetic data, −20.95, lies at the 72nd percentile of its distribution over noise realisations, so the realisation is typical; the three seeds reach −18.98 to −19.03 and meet criteria (a)–(e) (11 moves per free parameter, spread of the maximum log-likelihood 0.05, of the medians 0.05 posterior SD). The classification is reproduced entry by entry (Table S8, Fig. S3): the six entries identified on the measured data are identified again on the synthetic data, and the nine entries not identified on the measured data are not identified on the synthetic data either, although their true values are

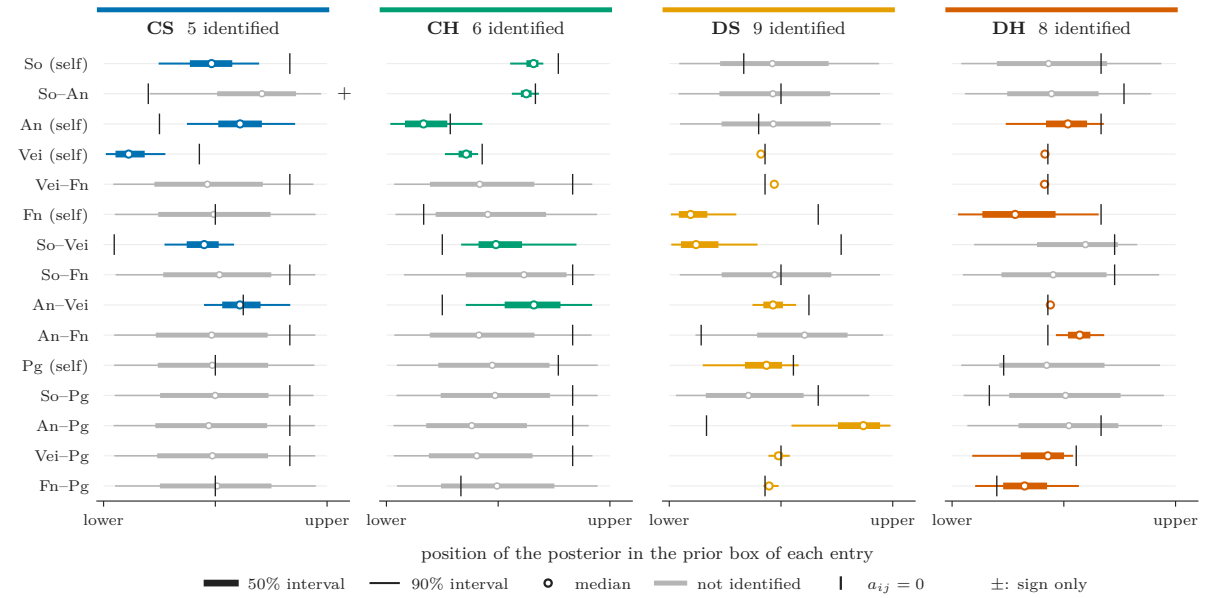


Figure S2: Marginal posteriors of the 15 entries of **A** at the final stage (three seeds pooled), drawn as position within the prior box of the final stage of each entry (Table S4): thick bar 50%, thin bar 90% interval, circle median; black tick: $a_{ij} = 0$. Coloured: identified; grey: not identified; a sign at the right: only the sign is determined (classification as in Fig. 3 of the main text). DH: stage (iii).

known and, for four of them, lie close to a bound of the box. Entries reported as not identified are therefore not identified even when the model is correct: this is a property of the data, not of the sampler. The point recovery of the identified entries is weaker: only two of the six 90% intervals contain the true value, and the posterior medians differ from the truth by up to 0.34 of the box width (a_{22}: truth +0.50, posterior −0.35 $[-0.92, +0.40]$; a_{33}: −0.71 against −4.5 $[-7.6, -1.5]$), while the sampler finds parameters that fit the noisy data better than the truth (−19.0 against −20.95). Over all 15 entries, 8 intervals contain the true value. Because the truth is a single point, the coverage of the 90% intervals is not an independent 0.9 trial per entry (entries are correlated, and Bayesian coverage is an average over the prior); but the under-coverage of the identified entries is consistent with the likelihood treating the species fractions as independent (main text, Section 4.5), and the posterior intervals of the identified entries should be read as lower bounds on their uncertainty. The same test on CS and DS (synthetic data from the best particle of the final-stage seed with the highest log-likelihood; 15000 and 54000 particles; criteria (a)–(e) met) gives the same picture with better point recovery: in CS all five identified entries are identified again and contain the true value (largest error 0.21 of the box width), and the nine not identified remain so; the one change of class is a_{12}, sign-only on the measured data and not identified on the synthetic data (its true value, +1.35, is recovered: median +1.24 $[-0.07, +1.93]$). In DS, eight of the nine identified entries are identified again and contain the true value (largest error 0.21), a_{13} becomes sign-only with the true value inside its interval, a_{25} (true +2.49 near the upper bound of the box) is identified but its interval $[+0.84, +2.46]$ just misses the truth, and the six not identified remain so; coverage over all 15 entries is 13/15 in both conditions. In DH (synthetic data from the best particle of the stage-(iii) seed with the highest log-likelihood; 60000 particles; criteria met) all eight identified entries are identified again with the true value inside the interval (largest error 0.13 of the box width), the seven not identified remain so, and all 15 intervals contain the truth; this realisation is, however, a favourable one (the log-likelihood of the true parameters, −18.75, lies at the 97th percentile of its distribution over noise realisations, against the 60th–72nd percentile in the other conditions).

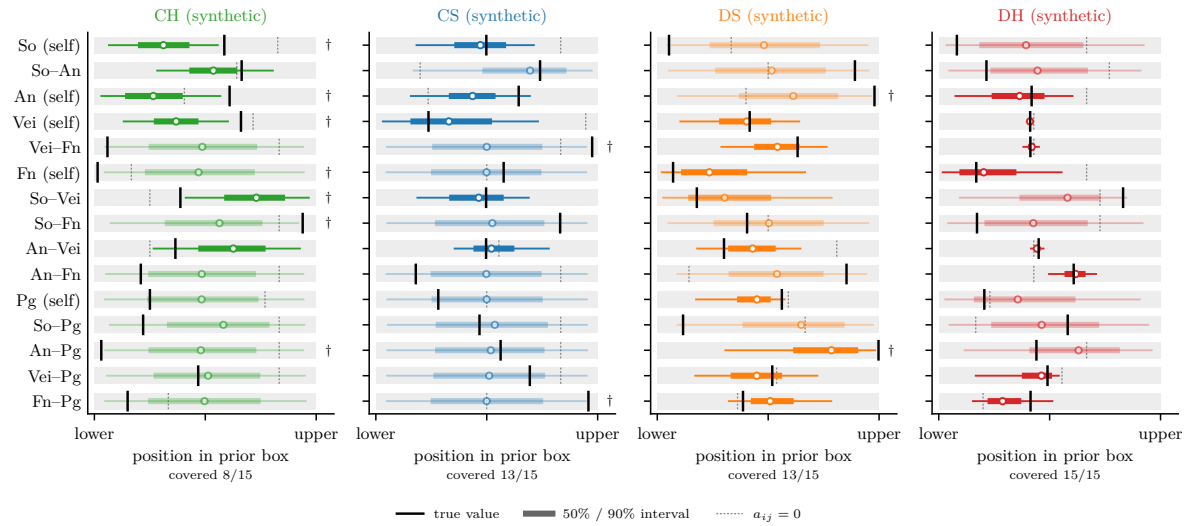


Figure S3: Recovery test: posterior on the synthetic data (thick bar 50%, thin bar 90% interval, circle median) and true value (black tick), drawn as position within the prior box of each entry; pale: not identified. †: interval does not contain the true value. Generated by `tools/recovery_evaluate.py`.

S8 MAP and posterior bulk by seed

Figure S4 overlays the MAP of each seed on the posterior predictive band pooled over the three seeds. In all conditions the three MAPs nearly coincide. The viable-composition error of the pointwise posterior median is 0.094 in CS, 0.086 in CH, 0.057 in DS and 0.083 in DH, against 0.090, 0.060, 0.057 and 0.073 at the MAP; in CH and DH the pointwise median, which is not itself a model trajectory, lies further from the data than the MAP.

S9 Bayes factor for the *Fn–Pg* interaction

For the Bayes factor, stage (i) was run for both models (full and $a_{45} = 0$) with 120 instead of 40 mutation steps, three seeds each, so that every group meets the acceptance criteria (a)–(e) (8.7–10.2 moves per free parameter in CS and CH; the exploratory stage-(i) groups with 40 steps reach only 2.7–3.6). Table S9 reports $\ln \hat{Z}$ of both models, $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$, and the width of the a_{45} box, which sets the Occam penalty.

Dysbiotic conditions. In DS and DH the data favour the full model ($\ln B = +7.5$ and $+2.1$), by more than the seed spread of $\ln \hat{Z}$ (up to 2.3 and 0.8). With the composition alone (stage (i)) a_{45} is positive in both: $+3.1$ to $+3.2$ (90% interval about $[+1.8, +5.1]$) in DS on the box $[-15, 20]$, and $+5.2$ in DH on the box $[-0.5, 6]$, where 22–26% of the posterior mass lies in the upper 5% of the box, so $\ln \hat{Z}_{\text{full}}$ of DH would change with a wider box (the Occam penalty grows with the box, the sign of $\ln B$ is unlikely to). At the final stage, with the live fraction in the likelihood, the 90% intervals of a_{45} include zero in both conditions (main text, Section 4.3). In the measured-data runs the stage-(i) posterior of DS has a single mode in a_{33} (no particle below -5 in any seed), unlike the read-off data used in an earlier version.

S10 Effect of the likelihood weights

The weights of the likelihood (main text, Section 3) encode two choices: the composition terms up-weight Pg ($\lambda_{\text{Pg}} = 5$), whose late increase is the main dynamic feature of the dysbiotic data,

and the last two time points ($\lambda_{\text{late}} = 3$), and down-weight species with a mean fraction below 5% ($\lambda_{\text{rare}} = 0.1$); the live-fraction term is weighted by λ_ψ (Table S2). To see how much the fit depends on these choices, the two-channel log-likelihood was maximised for each condition with three sets of weights: the current ones (A), no weights ($\lambda_{\text{Pg}} = \lambda_{\text{late}} = \lambda_{\text{rare}} = 1$, B) and $\lambda_\psi = 2$ in all conditions (C). Each maximisation starts from the MAP of stage (iii) (the seed with the highest log-likelihood) and uses the stage-(iii) box (L-BFGS-B, at most 150 iterations, measured data); it is a deterministic check of where the optimum moves, not a repeated sampling. Table S10 lists the fit of the three channels, a_{45} and the Pg ratio of day 21 to day 15 at the optimum (viable fractions, as in the main text). The weights change the error of each channel by at most 0.04 within a condition. In CS the live-fraction error stays at 0.23 under all three sets, so the conflict between composition and live fraction (Section S14) is not caused by the weights. The sign of a_{45} at the optimum does not change with the weights in CS, CH and DH (positive in CS and DH, negative in CH); in DS it does (+0.24 with the current weights, −0.93 without), consistent with a_{45} not being sign-determined there. In the commensal conditions a_{45} is not identified (its posterior spans the box), so its value at the optimum carries little information. In DH the up-weighting of Pg and of the late time points matters for the late Pg increase: the day-21/day-15 ratio is 16.9 with the current weights and 15.8 with $\lambda_\psi = 2$, against 14.9 measured, but 24.1 without weights; in DS, which shows no comparable late increase, the ratio stays at 1.5–1.6 under all three sets.

S11 Measured data and digitisation error

The composition, live-fraction and volume data were provided by the authors of Heine et al. (2025) as measured values: qRT-PCR species fractions with and without PMA for three to five biological replicates per condition and time point (CS 4, CH 5, DS 5, DH 3 from day 3 on), each measured in technical triplicate, and the live fraction and biofilm volume for each of 15–45 confocal images per time point. The technical triplicates scatter much less than the biological replicates (median ratio of the SDs 0.1–0.3 for the viable fractions), so the replicate spread used for σ_i reflects biological variability. An earlier version of this work used values read off the published figures, with replicate values generated from the box-plot quantiles. Table S11 compares the two: the live fraction and, except for DS, the biofilm volume were read off accurately, but the species fractions differ by up to 0.42, mostly in the dysbiotic conditions (e.g. *Veillonella* in DH at day 10, and the total-count fractions, which are plotted as stacked bars). The SDs derived from the generated replicates were also larger than the measured ones in the commensal conditions (for So 0.21 against 0.12 on average). All results reported here use the measured values (Fig. S5).

S12 Viable and total-count composition

The species fractions used in the inference are viable-cell fractions from quantitative real-time PCR with propidium monoazide (PMA) pre-treatment (Fig. 3 of Heine et al., 2025; medians over the measured replicates, Section S11). The corresponding total-count fractions without PMA (their Fig. S1) differ from them by an RMSE between the two sets of medians of 0.101 (CS), 0.087 (CH), 0.119 (DS) and 0.087 (DH). Because the species differ in viability, the two compositions enter the likelihood as separate channels (Eq. (5) of the main text): the viable fractions against the living fractions $\phi_i\psi_i / \sum_j \phi_j\psi_j$ and the total-count fractions against the volume fractions $\phi_i / \sum_j \phi_j$; the community live fraction from SYTO9/propidium iodide (LIVE/DEAD BacLight) staining, which is image based and not resolved by species, is the third term. Table S12 gives the errors of the final MAP in both channels and, for comparison, of the viable data against the volume fractions, i.e. the error a likelihood with a species-independent viability would see.

Part of the total-count error in the commensal conditions comes from species whose measured total-count fraction is zero or nearly so (Fn and Pg). The barrier penalty of Section S1, which grows as ϕ_i^{-3} in the stationarity condition, does not let a volume fraction fall below a floor: the total-count fractions of Fn and Pg in the model settle within the first days at 0.04–0.05 in CH and 0.02 in CS (Pg rising from $\phi_{\mathrm{Pg}}(0) = 10^{-3}$), although in CH the interactions of So, An and Vei with Pg sit at or near the lower edge of the box. The model keeps these species mostly dead instead: from day 6 on ψ_i is 0.05–0.06 for both species in CH and for Pg in CS, while Fn in CS dies more gradually (0.21 on day 6, 0.08 on day 21), so their viable fractions $\phi_i\psi_i/\sum_j \phi_j\psi_j$ stay below 0.007 and the viable composition is matched, whereas a residual of 0.02 (CS) to 0.05 (CH) per species remains in the total-count composition. With K_p lowered from 10^{-4} to 10^{-6} (same parameters, CH), ϕ_{Pg} on day 21 falls from 0.037 to 0.004, which identifies the floor as an effect of the barrier.

S13 Species-resolved viability implied by the data

The viable and the total-count fractions together imply a viability for each species, $\psi_i^{\mathrm{data}} = (y_i^{\mathrm{viable}}/y_i^{\mathrm{total}})\,\ell$, where y_i are the normalised fractions and ℓ is the community live fraction (medians over the measured replicates and mean over the images; days with a total-count fraction below 2% are left out). Resampling the biological replicates of both compositions and the biofilms of the live fraction (five images per biofilm; bootstrap) gives wide intervals: the 90% interval of ψ_i^{data} on a single day has a median width of 0.57 and its upper end exceeds 1 for 45% of the species and days, so ψ_i^{data} enters only stages (i)–(ii) as fixed viabilities and serves as a check, not as data in the likelihood. Table S13 compares it with the viability ψ_i of the final MAP. With the viable and the total-count compositions as separate channels of the likelihood, ψ_i is constrained species by species, but only through the products $\phi_i\psi_i$ of the viable composition and through the community live fraction; for a species that makes up a small part of the community both constrain ψ_i weakly. In CH the model reproduces the viability of the dominant species So (0.37 against 0.35 in the data), while An and Vei stay near 0.9 (data 0.24 and 0.53, whose single-day intervals are wide; see above). In CS the model viabilities of So, An and Vei are 0.96–0.97 against 0.50–0.80 in the data; this is the tension of CS described in Section 4.5 of the main text and in Section S14: the model has no term that keeps part of a growing population dead, so the community it needs for the compositions is almost fully viable. In both commensal conditions Fn and Pg are mostly dead in the model (0.05–0.11), in line with the one value available for Fn in CS (0.09); on the other days and for Pg their total-count fraction is below 2% and ψ_i^{data} is not defined.

S14 Composition against live fraction in CS

To test whether the CS tension depends on the weights, we minimised the negative log-likelihood of the main text (measured data, two compositions and the live fraction, biofilm-level σ_ℓ) with L-BFGS-B, started from the previous MAP, for λ_ψ = 1.5, 3, 6, 12 and 24 (Fig. S6). The errors move along a single curve: no weight brings both the composition and the live-fraction errors below 0.1. The same holds with the initial state taken from the day-1 data (total-count composition, species viabilities from the data; live-fraction error 0.237 against 0.231) and with a day-specific σ_ℓ (the between-biofilm SD of each day, at least 0.03), which moves CS along the same curve (live fraction 0.18, viable composition 0.12, total-count composition 0.09). In CH the same procedure gives errors of 0.07 (viable), 0.07 (total) and 0.12 (live fraction) at the default weight. A death term does not remove the tension either. Without antimicrobial agents the term $\alpha^* b_i\psi_i$ of Eq. (S8) is inactive; switching it on ($\alpha^* = 1$, $b_i \in [0, 10]$ added to the parameters) lowers the CS live-fraction error only from 0.23 to 0.22, although it brings the model viabilities closer to those implied by the data (error 0.36 to 0.27; in CH 0.45 to 0.25, where the live-fraction

error falls from 0.12 to 0.08). The death rates do not vanish in the flow-reactor condition CH, so the data do not single out the static cultures as the ones that need such a term. Whether the extra parameters are supported by the data would require a model comparison (Bayes factor), which we leave to future work; all results in the main text use $\alpha^* = 0$.

S15 Composition-only fit in CS

To separate a limitation of the model from the conflict between the composition and live-fraction channels in CS, we minimised the negative log-likelihood of the main text with the live-fraction channel switched off ($\lambda_\psi = 0$; viable and total-count compositions kept, measured data), with L-BFGS-B started from the MAP and from two random perturbations of it, and with every free entry restricted to $[-15, 20]$. The best solution reaches a viable-composition error of 0.078 and a total-count error of 0.042, against 0.090 and 0.039 at the MAP; the three starts end at 0.078–0.082. Its live-fraction error rises from 0.23 to 0.28. Dropping the live fraction altogether therefore improves the viable composition by only about 0.01: the remaining composition error in CS is a limit of the model, not a cost of fitting the live fraction. This is a deterministic diagnostic, not a posterior; its solution lies near the edge of the widened range ($a_{13} = 18.7$). Generated by `tools/two_channel_pilot_v2.py` (options in the archived output `CS_componly.json`).

S16 Growth-induced stress at a constrained material point

To show how the calibration enters a mechanical model, the posterior dynamics of each condition drive the isotropic growth law of the continuum model of Klempt et al. (2024): $\mathbf{F}_g = (1+\alpha)\mathbf{I}$ with $\mathrm{d}\alpha/\mathrm{d}T^* = k_\alpha \varphi_g$, where T^* is time normalised by the length of the experiment (21 days) and $k_\alpha = 10^{-3}$ as in that work. At a material point that is constrained in all directions, $\mathbf{F} = \mathbf{I}$, the elastic part is $\mathbf{F}_e = \mathbf{I}/(1+\alpha)$ with $J_e = (1+\alpha)^{-3}$, the deviatoric stress vanishes and the Cauchy stress is $\boldsymbol{\sigma} = K(J_e - 1)\mathbf{I}$, with $K = E/[3(1-2\nu)]$, $E = E_0(\varphi_{\mathrm{tot}}^2 + f)$, $\nu = 0.49$ and $f = 10^{-3}$, so no finite-element solution is needed. We report the growth pressure $p = -\frac{1}{3}\operatorname{tr}\boldsymbol{\sigma}$ relative to E_0 for two choices of the driver φ_g: the living volume $\sum_i \varphi_i \psi_i$, whose community fraction is constrained by the live/dead data, and the total volume $\varphi_{\mathrm{tot}} = \sum_i \varphi_i$, which the likelihood does not constrain (it uses fractions only). Driven by the total volume, the conditions are hardly distinguishable: φ_{tot} stays at 0.91–0.98 and p/E_0 on day 21 is 0.038–0.046 (DS, CH, DH, CS; Fig. S7b, dashed). Driven by the living volume, they separate: p/E_0 on day 21 is 0.023 (90% interval $[0.020, 0.025]$) in CH, 0.024 ($[0.023, 0.025]$) in DH, 0.021 ($[0.020, 0.022]$) in DS and 0.041 ($[0.038, 0.043]$) in CS. The high CS value follows the CS live fraction of the model (0.93 on day 21), which lies above the measured 0.65–0.77 (Section S14). This is an illustration of the coupling, not a prediction of the stress in a biofilm: k_α and E_0 are taken from another system and p scales linearly with both, and a spatial model adds nutrient transport, boundary conditions and the spatial distribution of the species. It shows that the dependence of the growth stress on the condition enters through the calibrated viability rather than through assumed species-specific growth parameters, and that the posterior spread can be propagated to the stress.

S17 Prediction of a held-out day

To check how well the calibrated model predicts data it has not seen, we repeated stages (i)–(iii) of CH with one observation day removed from all channels of the likelihood (day 10 or day 15; one seed, same boxes, weights and particle numbers as the main runs) and compared the posterior prediction for the removed day with the data. For the remaining days the weight λ_{late} falls on the last two of them (days 15 and 21 without day 10, days 10 and 21 without day 15). Both runs

meet criteria (a)–(c) (10.9 and 11.1 moves per free parameter); (d) and (e) need several seeds and are not evaluated. As a reference, the same day is evaluated for the stage-(iii) posterior of the main runs, which uses it (three seeds pooled). Table S15 gives the RMSE of the posterior median (300 posterior samples) and the number of species whose measured fraction lies in the central 90% predictive interval.

The viable composition of the removed day is predicted about as well as it is fitted when the day is used: the RMSE is 0.027 against 0.034 on day 10 and 0.041 against 0.022 on day 15. In both cases the two species outside the interval are Fn and Pg, whose measured viable fraction is zero while the model keeps them at about 0.005 (Section S12). The total-count composition and the community live fraction are not predicted better than they are fitted, but removing a day does not degrade them either: the total-count RMSE is the same to within 0.001, and the live-fraction error grows from 0.14 to 0.17–0.19; the live fraction lies outside the 90% interval in all four cases, with or without the day (Section S14). The total-count errors are dominated by Vei, which the model underestimates on both days, An (day 15), and the floor of Fn and Pg described in Section S12, none of which depends on whether the day is used.

References

Gelman, A., Rubin, D.B., 1992. Inference from iterative simulation using multiple sequences. Statistical Science 7, 457–472. doi:10.1214/ss/1177011136.

Hackl, K., 1997. Generalized standard media and variational principles in classical and finite elastoplasticity. Journal of the Mechanics and Physics of Solids 45, 667–688. doi:10.1016/S0022-5096(96)00110-X.

Heine, N., Bittroff, K., Szafrański, S.P., Duitscher, M., Behrens, W., Vollmer, C., Mikolai, C., Kommerein, N., Debener, N., Frings, K., Heisterkamp, A., Scheper, T., Torres-Mapa, M.L., Bahnemann, J., Stiesch, M., Doll-Nikutta, K., 2025. Influence of species composition and cultivation condition on peri-implant biofilm dysbiosis in vitro. Frontiers in Oral Health 6, 1649419. doi:10.3389/froh.2025.1649419.

Junker, P., Balzani, D., 2021. An extended Hamilton principle as unifying theory for coupled problems and dissipative microstructure evolution. Continuum Mechanics and Thermodynamics 33, 1931–1956. doi:10.1007/s00161-021-01017-z.

Junker, P., Bode, T., Hackl, K., 2025. On a holistic variational formulation for material modeling including dissipative evolution. Journal of the Mechanics and Physics of Solids 200, 106133. doi:10.1016/j.jmps.2025.106133.

Junker, P., Makowski, J., Hackl, K., 2014. The principle of the minimum of the dissipation potential for non-isothermal processes. Continuum Mechanics and Thermodynamics 26, 259–268. doi:10.1007/s00161-013-0299-4.

Klempt, F., Geisler, H., Soleimani, M., Junker, P., 2026. A continuum multi-species bacterial growth model with a novel interaction scheme. Archive of Applied Mechanics 96, 164. doi:10.1007/s00419-026-03160-y.

Klempt, F., Soleimani, M., Wriggers, P., Junker, P., 2024. A Hamilton principle-based model for diffusion-driven biofilm growth. Biomechanics and Modeling in Mechanobiology 23, 2091–2113. doi:10.1007/s10237-024-01883-x.

Kolenbrander, P.E., Palmer Jr., R.J., Periasamy, S., Jakubovics, N.S., 2010. Oral multispecies biofilm development and the key role of cell–cell distance. Nature Reviews Microbiology 8, 471–480.

Roberts, G.O., Rosenthal, J.S., 2001. Optimal scaling for various Metropolis–Hastings algorithms. Statistical Science 16, 351–367. doi:10.1214/ss/1015346320.

Vehtari, A., Gelman, A., Simpson, D., Carpenter, B., Bürkner, P.C., 2021. Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC. Bayesian Analysis 16, 667–718. doi:10.1214/20-BA1221.

Table S8: Recovery test (four conditions): true value, posterior median and 90% interval on the synthetic data, and classification on the synthetic and on the measured data (id., identified; s.o., sign only; n.i., not identified; italic: the two differ). †: interval does not contain the true value. Generated by `tools/recovery_evaluate.py`.

Entry	CH			CS		
	truth	median [5%, 95%]	synth./real	truth	median [5%, 95%]	synth./real
a_{11}	−0.58	−1.25 [−1.85, −0.66]†	id./id.	−4.02	−4.34 [−7.80, −1.45]	id./id.
a_{12}	+0.05	−0.23 [−0.79, +0.36]	id./id.	+1.35	+1.24 [−0.07, +1.93]	*n.i./s.o.*
a_{22}	+0.50	−0.35 [−0.92, +0.40]†	id./id.	+1.60	+0.79 [−0.31, +1.80]	id./id.
a_{33}	−0.71	−4.52 [−7.59, −1.48]†	id./id.	−11.24	−9.78 [−14.50, −3.42]	id./id.
a_{34}	−4.65	−2.08 [−4.70, +0.65]	n.i./n.i.	+0.85	−2.00 [−4.70, +0.68]†	n.i./n.i.
a_{44}	−0.91	+1.82 [−0.72, +4.66]†	n.i./n.i.	+0.15	−0.00 [−0.90, +0.90]	n.i./n.i.
a_{13}	+0.55	+1.92 [+0.65, +2.87]†	id./id.	+8.28	+7.75 [+3.34, +11.35]	id./id.
a_{14}	+0.63	−1.62 [−4.56, +0.51]†	n.i./n.i.	−0.02	−1.85 [−4.69, +0.70]	n.i./n.i.
a_{23}	+0.46	+1.50 [+0.07, +2.71]	id./id.	−0.39	−0.22 [−1.34, +1.52]	id./id.
a_{24}	−3.74	−2.09 [−4.68, +0.65]	n.i./n.i.	−3.92	−2.01 [−4.71, +0.71]	n.i./n.i.
a_{55}	−3.37	−1.86 [−4.69, +1.13]	n.i./n.i.	−0.44	−0.00 [−0.90, +0.90]	n.i./n.i.
a_{15}	−3.68	−1.51 [−4.58, +0.68]	n.i./n.i.	−2.20	−1.78 [−4.67, +0.71]	n.i./n.i.
a_{25}	−4.81	−2.12 [−4.71, +0.65]†	n.i./n.i.	−1.62	−1.89 [−4.69, +0.72]	n.i./n.i.
a_{35}	−2.19	−1.92 [−4.66, +0.70]	n.i./n.i.	−0.83	−1.93 [−4.70, +0.70]	n.i./n.i.
a_{45}	−0.55	+0.49 [−0.85, +1.86]	n.i./n.i.	+0.92	−0.00 [−0.90, +0.89]†	n.i./n.i.
	covered 8/15 (identified 2/6); re-identified 6/6			covered 13/15 (identified 5/5); re-identified 5/5		

Entry	DS			DH		
	truth	median [5%, 95%]	synth./real	truth	median [5%, 95%]	synth./real
a_{11}	−0.84	+0.44 [−0.86, +1.84]	n.i./n.i.	−8.77	−4.09 [−9.47, +3.86]	n.i./n.i.
a_{12}	+0.78	+0.03 [−0.89, +0.90]	n.i./n.i.	−7.20	−4.21 [−9.38, +1.83]	n.i./n.i.
a_{22}	+1.45	+0.53 [−0.77, +1.41]†	n.i./n.i.	−3.72	−4.53 [−8.87, −0.96]	id./id.
a_{33}	−0.73	−0.75 [−1.09, −0.48]	id./id.	−0.57	−0.60 [−0.89, −0.47]	id./id.
a_{34}	+1.72	+1.50 [+0.89, +2.03]	id./id.	−0.54	−0.34 [−1.64, +0.81]	id./id.
a_{44}	−4.73	−4.11 [−4.92, −2.48]	id./id.	−7.46	−6.96 [−9.73, −1.69]	id./id.
a_{13}	−4.19	−3.62 [−4.88, −1.43]	*s.o./id.*	+1.14	−1.61 [−6.95, +1.30]	n.i./n.i.
a_{14}	−0.19	+0.01 [−0.90, +0.90]	n.i./n.i.	−6.10	−3.31 [−7.53, +2.11]	n.i./n.i.
a_{23}	−1.96	−1.46 [−2.42, −0.63]	id./id.	+0.77	+0.49 [−0.44, +1.52]	id./id.
a_{24}	+2.49	+1.39 [−0.18, +2.80]	n.i./n.i.	+6.30	+6.62 [+2.41, +9.82]	id./id.
a_{55}	−0.25	−1.20 [−3.52, −0.14]	id./id.	−0.16	+0.82 [−1.31, +4.39]	n.i./n.i.
a_{15}	−4.13	−0.13 [−4.31, +2.29]	n.i./n.i.	+1.24	+0.89 [−0.35, +2.33]	n.i./n.i.
a_{25}	+2.49	+1.98 [+0.84, +2.46]†	id./id.	−1.70	−0.27 [−4.12, +2.20]	n.i./n.i.
a_{35}	−0.03	−0.14 [−0.57, +0.29]	id./id.	−2.93	−4.16 [−17.50, −0.64]	id./id.
a_{45}	+0.15	+0.92 [−0.24, +2.65]	id./id.	+5.35	+2.20 [−1.15, +7.81]	id./id.
	covered 13/15 (identified 8/9); re-identified 8/9			covered 15/15 (identified 8/8); re-identified 8/8		

Table S9: Log-evidence of the full and the reduced ($a_{45} = 0$) model at stage (i), 120 mutation steps, mean ± half-range over three seeds, and $\ln B = \ln \hat{Z}_{\text{full}} - \ln \hat{Z}_{a_{45}=0}$. The last two columns give the a_{45} box and its log-width, which enters $\ln \hat{Z}_{\text{full}}$ as an Occam penalty.

Cond.	$\ln \hat{Z}_{\text{full}}$	$\ln \hat{Z}_{a_{45}=0}$	$\ln B$	a_{45} box	ln(width)
CS	-16.58 ± 0.13	-16.52 ± 0.03	−0.06	[-1, 1]	0.69
CH	-13.18 ± 0.05	-13.21 ± 0.08	+0.03	[-1, 2]	1.10
DS	-87.31 ± 0.94	-94.76 ± 1.15	+7.46	[-15, 20]	3.56
DH	-35.15 ± 0.42	-37.22 ± 0.22	+2.07	[-0.5, 6]	1.87

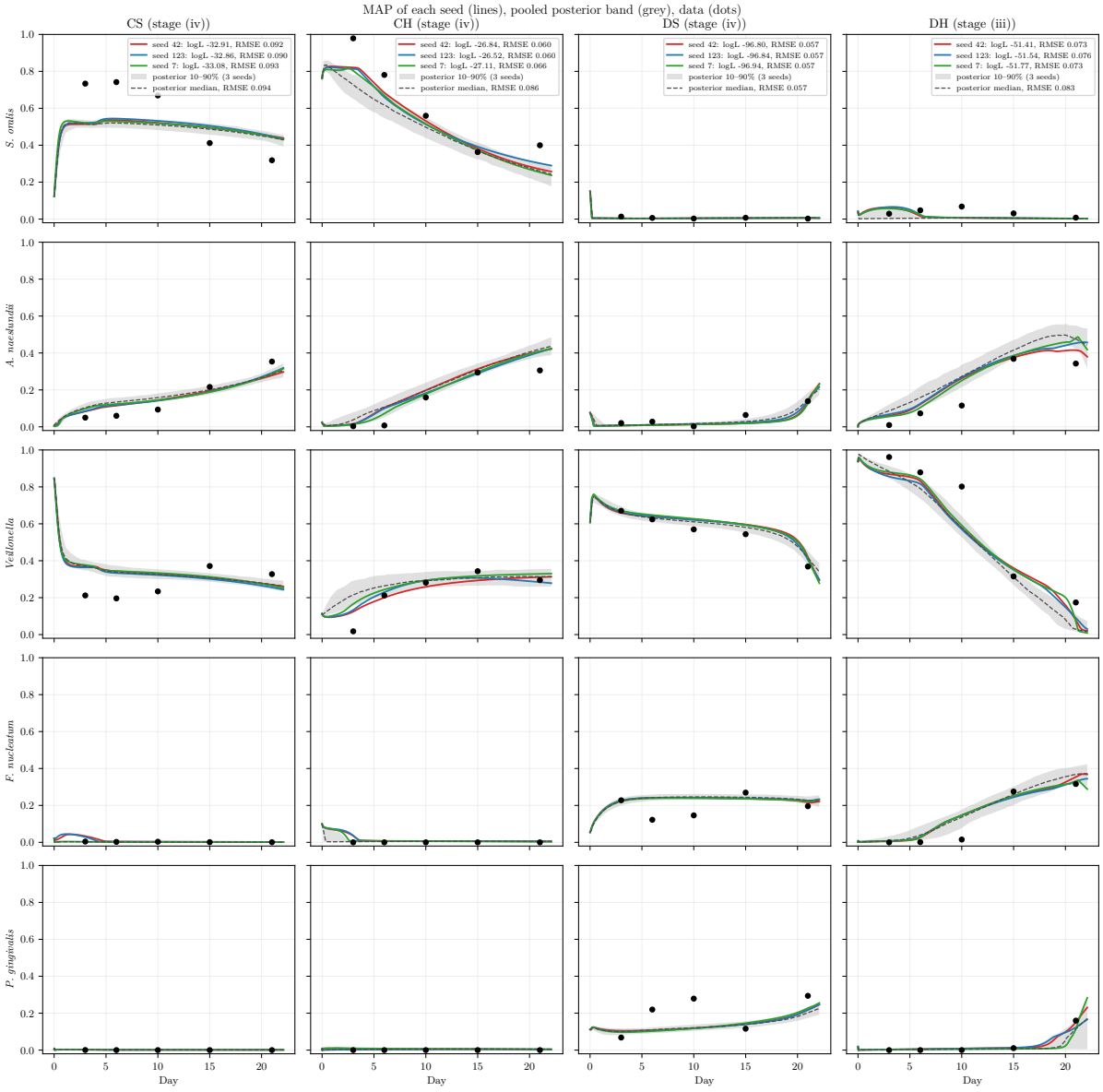


Figure S4: Species fractions: MAP of each seed (coloured lines), pooled central 80% posterior predictive band (10th–90th percentile, grey) and its pointwise median (dashed), and data (dots). DH: stage (iii).

Table S10: Two-channel fit with the current weights (A), without weights (B) and with $\lambda_\psi = 2$ in all conditions (C), at the maximum of the respective log-likelihood. RMSE viable: viable fractions against $\phi_i\psi_i / \sum_j \phi_j\psi_j$; RMSE total: total-count fractions against $\phi_i / \sum_j \phi_j$; RMSE live: community live fraction. Generated by `tools/weight_sens_table.py`.

	Weights	RMSE viable	RMSE total	RMSE live	a_{45}	Pg D21/D15
CS	current	0.091	0.039	0.232	+0.90	0.97
	unweighted	0.086	0.031	0.235	+0.87	0.98
	$\lambda_\psi = 2$	0.092	0.041	0.231	+0.91	0.94
CH	current	0.059	0.078	0.121	−0.10	0.78
	unweighted	0.063	0.088	0.081	−0.52	0.88
	$\lambda_\psi = 2$	0.059	0.082	0.105	−0.02	0.81
DS	current	0.058	0.080	0.094	+0.24	1.56
	unweighted	0.054	0.079	0.091	−0.93	1.50
	$\lambda_\psi = 2$	0.058	0.080	0.094	+0.24	1.56
DH	current	0.069	0.156	0.077	+5.65	16.92
	unweighted	0.064	0.168	0.057	+6.73	24.13
	$\lambda_\psi = 2$	0.069	0.151	0.092	+5.77	15.80

Table S11: Values read off the published figures against the measured values: largest absolute difference over days and species (RMSE in parentheses). Volume: relative difference of the median. Generated by `tools/raw_data_report.py`.

	CS	CH	DS	DH
Viable fractions (PMA)	0.06 (0.02)	0.09 (0.03)	0.17 (0.05)	0.42 (0.11)
Total-count fractions	0.22 (0.08)	0.23 (0.09)	0.37 (0.15)	0.42 (0.15)
Live fraction	0.009 (0.006)	0.015 (0.008)	0.011 (0.006)	0.010 (0.006)
Biofilm volume (rel.)	0.03 (0.02)	0.14 (0.08)	0.57 (0.25)	0.03 (0.03)

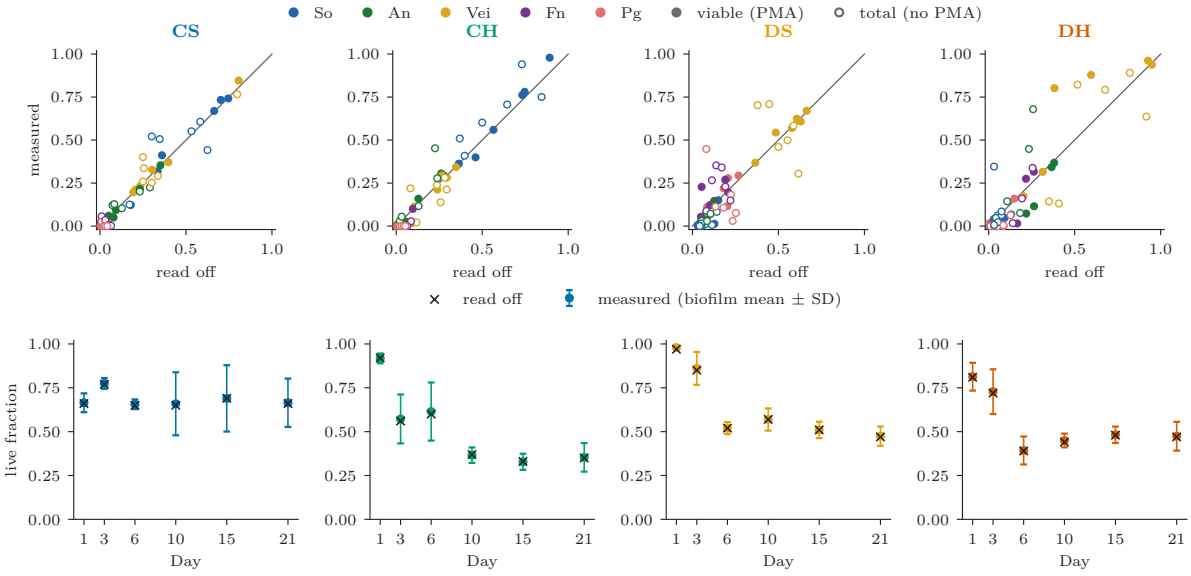


Figure S5: Values read off the published figures against the measured values. Top: species fractions (median over replicates per day and species; filled, viable cells with PMA; open, all cells without PMA; line, identity). Bottom: community live fraction, measured (mean ± SD over biofilms, five images per biofilm) and read off (crosses). Generated by `tools/fig_raw_vs_digitised.py`.

Table S12: RMSE of the MAP of the final stage (seed with the highest maximum log-likelihood) against the medians of the viable-cell data (PMA-qRT-PCR) and of the total-count data (Heine et al., 2025, Fig. S1). The first two rows are the channels of the likelihood; the last row compares the viable data with the volume fractions and is not used in the fit.

	CS	CH	DS	DH
Viable vs. $\phi_i\psi_i / \sum_j \phi_j\psi_j$	0.090	0.060	0.057	0.073
Total count vs. $\phi_i / \sum_j \phi_j$	0.039	0.081	0.081	0.159
Viable vs. $\phi_i / \sum_j \phi_j$ (not fitted)	0.110	0.107	0.082	0.105

Table S13: Viability of each species: implied by the data, ψ_i^{data} (median over days 3–21), and of the final MAP, ψ_i (median over the same days). Data: `tools/psi_from_data.py`; model: MAP of the seed with the highest maximum log-likelihood, `tools/psi_table_runs.py`; –: total-count fraction below 2% on every day.

	So		An		Vei		Fn		Pg	
	data	MAP	data	MAP	data	MAP	data	MAP	data	MAP
CS	0.80	0.97	0.59	0.96	0.50	0.96	0.09	0.11	–	0.06
CH	0.35	0.37	0.24	0.88	0.53	0.89	–	0.06	–	0.05
DS	–	0.05	0.74	0.09	0.65	0.57	0.60	0.75	1.06	0.68
DH	0.35	0.06	0.26	0.91	0.58	0.36	0.63	0.88	1.19	0.07

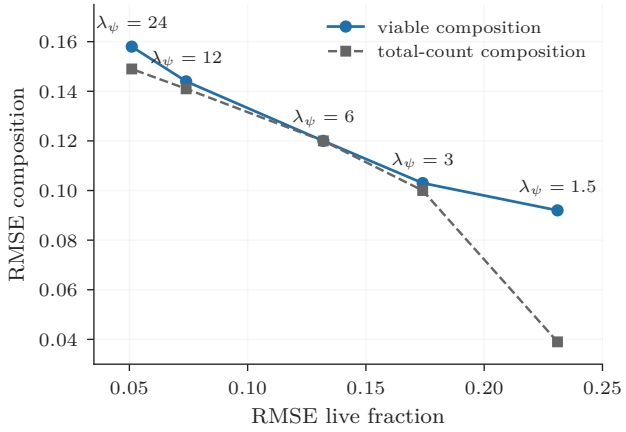


Figure S6: CS: error of the viable (circles) and total-count (squares) composition against the error of the live fraction when the live-fraction weight λ_ψ is varied (deterministic optimisation, measured data). Generated by `tools/two_channel_pilot_v2.py`.

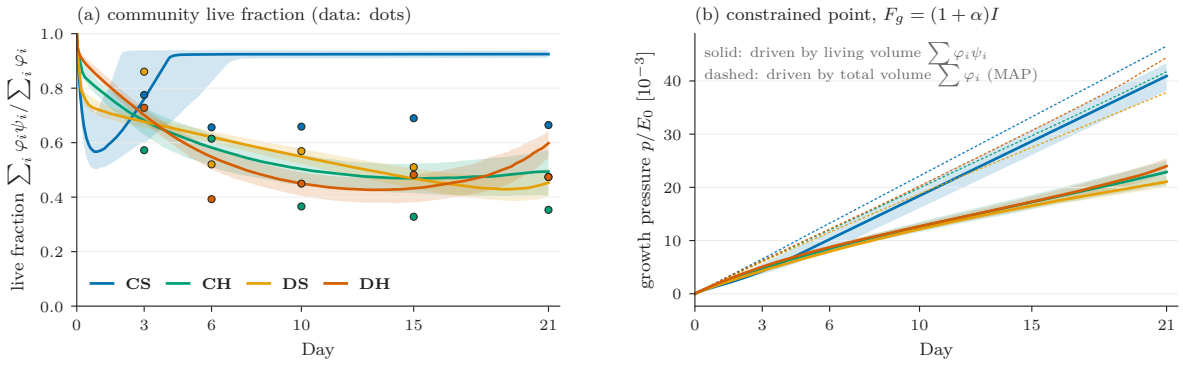


Figure S7: Growth-induced stress at a constrained material point. (a) Community live fraction $\sum_i \varphi_i\psi_i / \sum_i \varphi_i$ (line: posterior median, band: central 90%, 200 posterior samples per condition; dots: live/dead data used in the inference). (b) Growth pressure p/E_0 for growth driven by the living volume (solid, with band) and by the total volume (dashed, MAP). Generated by `tools/growth_stress_check.py`. DH: stage (iii).

Cond.	Stage	N_p	n_{mut}	K	Wall time / seed	N_{lik}	GPU
CS	(i)	1000	40	8	1.6 h	3.2×10^5	RTX 4090
	(ii)	2000	120	6	3.6 h	1.4×10^6	RTX 4090
	(iii)	2000	160	7	7.9 h	2.2×10^6	RTX 4090
	(iv)	5000	150	8	7.2 h	6.0×10^6	RTX 4090
CH	(i)	1000	40	8	1.8 h	3.2×10^5	RTX 4090
	(ii)	2000	150	6	4.9 h	1.8×10^6	RTX 4090
	(iii)	2000	150	8	6.6 h	2.4×10^6	RTX 4090
	(iv)	5000	150	9	7.8 h	6.8×10^6	RTX 4090
DS	(i)	1000	40	12	3.1 h	4.8×10^5	RTX 4090
	(ii)	3000	100	9	6.8 h	2.7×10^6	RTX 4090
	(iii)	18000	150	13	21.7 h	3.5×10^7	RTX 4090
	(iv)	18000	150	11	17.6 h	3.0×10^7	RTX 4090
DH	(i)	1000	40	11	2.9 h	4.4×10^5	RTX 4090
	(ii)	2000	80	9	5.2 h	1.4×10^6	RTX 4090
	(iii)	20000	120	12	21.5 h	2.9×10^7	RTX 4090
	(iv)	20000	150	11	21.7 h	3.3×10^7	RTX 4090

Table S14: Cost of each inference stage (median over the three seeds): particle number N_p, mutation steps per tempering step n_{mut}, number of tempering steps K, wall time per seed on one GPU, and number of likelihood evaluations $N_{\mathrm{lik}} = N_p(1 + \sum n_{\mathrm{mut}})$. DH: stage (iii) is the final stage; stage (iv) is listed for completeness. Generated by `tools/make_cost_table.py`.

Table S15: Prediction of a held-out day in CH: RMSE of the posterior median against the data of the removed day (in parentheses: species, of five, whose measured fraction lies in the central 90% predictive interval), and absolute error of the community live fraction. Held out: day removed from the likelihood (stages (i)–(iii), one seed); used: stage (iii) of the main runs (three seeds). 300 posterior samples. Generated by `tools/holdout_eval.py`.

Day	Day in likelihood	Viable composition	Total-count composition	Live fraction
10	held out	0.027 (3)	0.060 (2)	0.17
10	used	0.034 (3)	0.060 (2)	0.14
15	held out	0.041 (3)	0.072 (0)	0.19
15	used	0.022 (3)	0.072 (0)	0.14